

2526-MA378: Assignment 2 ANS with outline solutions

Information on the deadline, weighting, collaboration policy, submission process, etc, is on Canvas. Direct link: <https://universityofgalway.instructure.com/courses/46941/assignments/140160>

Please take care to present your work as neatly as possible: 1 (of 5) marks will be awarded quality of exposition. Collaboration is encouraged, but please list anybody you collaborated with in your solution set (makes grading easier for me!).

Q1. Suppose we have a set of equally spaced interpolation points $\{x_0, x_1, \dots, x_N\}$, with $h = x_i - x_{i-1}$ for all i . For a differentiable function f , we invoke the usual notation $f_i := f(x_i)$, $f'_i := f'(x_i)$. Set

$$B_0 = f_{i-1}, \quad B_1 = f_{i-1} + \frac{h}{3}f'_{i-1}, \quad B_2 = f_i - \frac{h}{3}f'_i, \quad B_3 = f_i.$$

Define

$$H(t) := B_0(1-t)^3 + 3B_1 t(1-t)^2 + 3B_2 t^2(1-t) + B_3 t^3.$$

Now let S be the piecewise polynomial on the above interpolation points, where the piece of S on the interval $[x_{i-1}, x_i]$ is denoted S_i , and defined $S_i(x) = H(t)$ where $t = (x - x_{i-1})/h$. Show that this is the PCHIP Spline interpolant to f .

Answer: There are a few possible approaches to this, but for each we need to verify the following:

- (a) S is a piecewise polynomial of degree 3;
- (b) $S_i(x_{i-1}) = f(x_{i-1})$ and/or $S_i(x_i) = f(x_i)$;
- (c) $S'_i(x_{i-1}) = f'(x_{i-1})$ and/or $S'_i(x_i) = f'(x_i)$.

For (a), we note that t is a linear function of x , so each of the expressions in H , being cubic in t , are also cubic in x . So $H(t)$, and consequently $S_i(x)$, are cubics. Thus S is piecewise cubic.

For (b), note that $S_i(x_{i-1}) = H(0) = B_0 = f_{i-1}$. Also $S_i(x_i) = H(1) = B_3 = f_i$, as required.

For (c); first differentiate H to get

$$H'(t) := -3B_0(1-t)^2 + 3B_1(1-4t+3t^3) + 3B_2(2t-3t^2) + 3B_3t^2.$$

Also (by the Chain Rule) $\frac{d}{dx} S(x) = (1/h) \frac{d}{dt} H(t)$. So $S'(x_{i-1}) = (1/h)H'(0) = -3B_0 + 3B_1 = (1/h)(-3f_{i-1} + 3f'_{i-1}h) = f'_{i-1}$. Similarly $S'(x_i) = (1/h)H'(1) = f'_i$.

Q2. Let $f(x) = x^{3/2}$ and let S be its PCHIP Spline interpolant at the nodes $\{x_i\}_{i=0}^2 = \{1, 2, 3\}$

- (a) Evaluate S at the point $x = 2.5$.
- (b) Compute the (actual) the error at $x = 2.5$.
- (c) Compare this result with the error bound provided in Theorem 3.2

Answer: (a) Note that you are not asked to write down the formula for $S(x)$, so we only need $S_1(x)$, which is the piece of S on $[2, 3]$. You can use the usual formulation (as the notes) or the version from Q1, as you prefer. If the latter, you should find that

$$c_0 = 2\sqrt{2}, \quad c_1 = (3/2)\sqrt{2}, \quad c_2 = (15/2)\sqrt{3} - 9\sqrt{2}, \quad c_3 = -(9/2)\sqrt{3} + (11/2)\sqrt{2}.$$

Then we get (with a little arithmetic) that $S(2.5) = (21/16)\sqrt{3} + (19/16)\sqrt{2} \approx 3.9527$.

For (b), we compute $|f(2.5) - S(2.5)| \approx 1.5178 \times 10^{-4}$.

(c) First compute $f^{(4)}(x) = (9/16)x^{-5/2}$. We then see that $\|f^{(4)}\|_\infty = f^{(4)}(1) = 9/16$. Since $h = 1$, the error is bounded as $(9/16)/384 = 3/2048 \approx 1.465 \times 10^{-3}$. (Actually, if you work through the details of the proof of Thm 3.2, you'll see that we can actually bound the error on $[x_{i-1}, x_i]$ as $(h^4/384) \max_{x_{i-1} \leq x \leq x_i} |f^{(4)}(x)|$. With that you'd get the sharper estimate of 2.5895×10^{-4}).

Q3. Let $Q_0(\cdot)$ be the quadrature rule that approximates $\int_a^b f(x)dx$ as $Q_0(f) := (b-a)f\left(\frac{a+b}{2}\right)$.

(a) What is the *precision* of Q_0 ? Explain your answer. How does it compare with the precision of the Trapezium and Simpson's Rules?

(b) Derive a *composite* version of $Q_0(\cdot)$.

In your opinion, does this have any advantages over the Trapezium Rule? Explain your answer. (Note: since you are asked for an opinion, there is no one correct answer. However, your explanation may, or may not, be correct).

Answer: (a) This has precision 1 (it is exact for all linear polynomials). So, the precision is the same as the Trapezium Rule, but lower than Simpson's, which has 3.

(b) The composite version of this can be written as

$$T_0(f) = h \sum_{i=0}^N f(x_i + h/2),$$

where we have N intervals, $h = (b-a)/N$ and $x_i = a + ih$.

One could argue that this is slightly more efficient than the Trapezium Rule since it has the same precision, but requires one less function evaluation. Equally, you could argue that it is not any better since it requires computing midpoints of intervals.

Q4. Consider the quadrature rule:

$$R(f) = q_0 f\left(\frac{1}{3}\right) - f\left(\frac{1}{2}\right) + q_2 f\left(\frac{3}{4}\right)$$

for approximating $\int_0^1 f(x)dx$.

(a) Determine values of q_0 and q_2 that ensure this rule has precision 2.

Answer: (a) We need to find q_0 and q_2 so that $R(f) = \int_0^1 p_2(x)dx$ where p_2 is any polynomial of degree 2. Since that space of polynomials is spanned by the set $\{1, x, x^2\}$, we take q_0 and q_2 to satisfy the equations $q_0 - 1 + q_2 = 1$, $q_0/2 - 1/2 + q_2(3/4) = 1/2$, and $q_0/9 - 1/4 + q_2(9/16) = 1/3$. These equations are not linearly independent (since there are only two unknowns). Solving any pair of them should give $q_0 = 6/5$ and $q_2 = 4/5$. So $R(f) = \frac{6}{5}f\left(\frac{1}{3}\right) - f\left(\frac{1}{2}\right) + \frac{4}{5}f\left(\frac{3}{4}\right)$.

(b) What is the maximum precision of $R(\cdot)$ with the values of q_1 and q_2 that you have determined?

Answer: (b) Could this method be exact for some higher degree polynomials? Checking with $f(x) = x^3$, we should find that $R(x^3) = 37/144 \neq \int_0^1 x^4 dx$. So the precision is at most 2.

(c) Why is this not, strictly speaking, a Newton-Cotes rule?

Answer: (c) Either one of the following reasons would suffice: the limits of integration are not included as quadrature points, and the points are not equally spaced.